

Databricks One-Day Practical Assignment

Grocery Basket Behaviour, Reorder Pattern, and Product Strategy

Duration	One day
Mode	Individual practical submission
Platform	Databricks notebook
Dataset	Instacart Market Basket Analysis

Dataset Link

<https://www.kaggle.com/competitions/instacart-market-basket-analysis/data>

Use the files available in the dataset package. The dataset contains customer orders, product details, aisle and department information, and prior purchase behaviour from an online grocery platform.

Business Background

A grocery delivery company wants to understand how customers build their baskets, which products are frequently reordered, which departments drive repeated purchases, and where the business can improve product discovery and basket value. The company receives order-level and product-level data from its ordering platform and wants a Databricks-based analytical output that can help the product, operations, and growth teams make better decisions.

Problem Statement

You are asked to work as a data analyst on this grocery platform dataset. Your task is to use Databricks to convert the available raw files into a meaningful business analysis that explains customer ordering behaviour, product reorder patterns, department or aisle-level performance, and possible opportunities for improving repeat purchase and basket growth.

The business does not want a simple file reading notebook. It expects a practical Databricks notebook that shows how raw data can be understood, prepared, analysed, and presented as useful business output. You may decide your own approach, table design, transformations, validations, and final analysis flow.

Expected Work

Use Databricks as the main working environment. Apply any relevant concepts learned in the data analytics and Databricks classes, including Python, Spark DataFrames, SQL, data quality checks, joins, aggregation, Delta tables, reusable tables or views, and business-level interpretation. You are free to decide how deeply you want to use these concepts based on your own design.

Final Outcome

Submit one well-structured Databricks notebook or exported notebook that presents a complete analysis of the grocery basket business problem. The final output should clearly show what the business can learn from the data, which customer or product patterns are important, and what recommendations can be made from the analysis.

Submission Expectation

The notebook should be readable by a reviewer without verbal explanation. It should contain code, outputs, and short written observations wherever needed. The final section should summarize the business findings and recommendations in a clear manner.

Assessment Focus

Marks will be based on practical Databricks usage, correctness of data handling, depth of analysis, clarity of notebook structure, quality of business interpretation, and the student's ability to go beyond basic commands and think like a real data analyst.

Note: The solution approach is intentionally open. Students are expected to make their own design decisions and justify them through the notebook output.